

Data Engineering & Analytics Interview Cheat Sheet

This guide explains the terms from the diagram in a way that helps you answer confidently in interviews. Each section includes:

- Simple explanation
 - Interview-ready answer
 - Real-world example
 - Smart line to impress interviewer
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1. ETL (Extract, Transform, Load)

What it means

Data is extracted from source systems, transformed into the required format, and then loaded into a target database or warehouse.

Interview Answer

"ETL is mainly used when transformation logic is heavy and data quality checks must happen before storing data in the warehouse."

Example

An e-commerce company extracts sales data from MySQL, cleans invalid records, calculates revenue metrics, and loads the final data into Snowflake.

Impress Interviewer

"Traditional ETL is compute-intensive outside the warehouse, which was common before cloud-native scalable warehouses became popular."

2. ELT (Extract, Load, Transform)

What it means

Data is first loaded into the warehouse and transformed later using warehouse compute power.

Interview Answer

"ELT is preferred in modern cloud platforms like Snowflake and BigQuery because warehouses scale compute independently."

Example

Netflix loads raw streaming logs into BigQuery and transforms them later using SQL models.

Impress Interviewer

"ELT supports faster ingestion and enables reproducible transformations directly inside analytical systems."

3. Data Warehouse

What it means

A structured storage system optimized for analytics and reporting.

Interview Answer

"A data warehouse stores cleaned, structured, historical data optimized for BI dashboards and analytical queries."

Example

Amazon Redshift stores sales, customer, and inventory data for executive reporting.

Impress Interviewer

"Warehouses are schema-on-write systems, which improves query performance and governance."

4. Data Lake

What it means

A storage system that keeps raw structured, semi-structured, and unstructured data.

Interview Answer

"Data lakes allow organizations to store raw data cheaply before deciding how to use it."

Example

A company stores JSON logs, images, videos, and CSV files in AWS S3.

Impress Interviewer

"Data lakes follow schema-on-read, making them highly flexible for machine learning workloads."

5. Data Lakehouse

What it means

Combines the flexibility of data lakes with warehouse-like performance.

Interview Answer

"Lakehouse architecture provides ACID transactions and analytics directly on low-cost object storage."

Example

Databricks Delta Lake allows analytics on S3 data with warehouse-level reliability.

Impress Interviewer

"Lakehouses reduce data duplication because analytics and ML teams can work on the same storage layer."

6. Batch Processing

What it means

Processing large volumes of data at scheduled intervals.

Interview Answer

"Batch processing is ideal when near real-time insights are not required."

Example

A payroll system calculates employee salaries every night.

Impress Interviewer

"Batch jobs are generally more cost-efficient because resources can be optimized around scheduled workloads."

7. Stream Processing

What it means

Continuous processing of incoming data in real time.

Interview Answer

"Stream processing handles events immediately as they arrive."

Example

Uber processes live GPS events to estimate ride arrival times.

Impress Interviewer

"Frameworks like Kafka Streams and Apache Flink support stateful real-time processing."

8. Real-Time Processing

What it means

Immediate processing with minimal delay.

Interview Answer

"Real-time systems are used where latency directly affects business outcomes."

Example

Fraud detection in banking transactions.

Impress Interviewer

"Real-time architectures usually prioritize low latency over maximum throughput."

9. Data Pipeline

What it means

An automated system that moves data between systems.

Interview Answer

"A data pipeline automates ingestion, transformation, validation, and delivery of data."

Example

Moving customer data from PostgreSQL to Snowflake daily.

Impress Interviewer

"Reliable pipelines include retry mechanisms, monitoring, and lineage tracking."

10. Schema

What it means

Structure of data including fields, data types, and constraints.

Interview Answer

"A schema defines how data is organized and validated."

Example

Customer table with columns: customer_id, name, email, signup_date.

Impress Interviewer

"Good schema design improves query efficiency and maintainability."

11. Snowflake Schema

What it means

A normalized warehouse schema with multiple related tables.

Interview Answer

"Snowflake schema reduces redundancy by normalizing dimensions into sub-dimensions."

Example

A product dimension split into category and supplier tables.

Impress Interviewer

"Snowflake schemas save storage but can increase join complexity."

12. Star Schema

What it means

A central fact table connected to denormalized dimension tables.

Interview Answer

"Star schema is optimized for analytical query performance."

Example

Sales fact table linked to customer, product, and date dimensions.

Impress Interviewer

"BI tools perform efficiently on star schemas due to fewer joins."

13. Data Modeling

What it means

Designing how data entities relate to each other.

Interview Answer

"Data modeling ensures scalable, maintainable, and business-friendly analytics structures."

Example

Designing relationships between orders, customers, and products.

Impress Interviewer

"Good modeling balances normalization for integrity and denormalization for performance."

14. OLTP (Online Transaction Processing)

What it means

Systems optimized for fast transactional operations.

Interview Answer

"OLTP systems handle frequent inserts, updates, and deletes with high concurrency."

Example

Banking applications processing transactions.

Impress Interviewer

"OLTP databases prioritize consistency and low latency over analytical complexity."

15. OLAP (Online Analytical Processing)

What it means

Systems optimized for analytical queries and reporting.

Interview Answer

"OLAP systems aggregate large historical datasets for business intelligence."

Example

Sales trend dashboards in Power BI.

Impress Interviewer

"OLAP queries are typically read-heavy and involve aggregations across dimensions."

16. Distributed Systems

What it means

Multiple machines working together as one system.

Interview Answer

"Distributed systems improve scalability, fault tolerance, and parallel processing."

Example

Apache Hadoop cluster processing terabytes of logs.

Impress Interviewer

"Distributed systems trade complexity for horizontal scalability."

17. Message Queue

What it means

A system that stores and transfers messages asynchronously.

Interview Answer

"Message queues decouple producers and consumers for reliable communication."

Example

Kafka buffering clickstream events.

Impress Interviewer

"Queues improve resilience because producers and consumers can scale independently."

18. Data Partitioning

What it means

Splitting data into smaller segments.

Interview Answer

"Partitioning improves query performance and parallelism."

Example

Partitioning sales data by transaction_date.

Impress Interviewer

"Good partition keys reduce unnecessary data scans."

19. Data Sharing

What it means

Sharing datasets securely across systems or teams.

Interview Answer

"Data sharing eliminates duplicate copies while maintaining governance."

Example

Snowflake Secure Data Sharing between departments.

Impress Interviewer

"Modern platforms support zero-copy sharing for efficiency."

20. Indexing

What it means

Creating lookup structures for faster queries.

Interview Answer

"Indexes improve retrieval speed but add storage and write overhead."

Example

Index on customer_id for quick searches.

Impress Interviewer

"Choosing the wrong index strategy can degrade insert performance."

21. Data Governance

What it means

Policies and controls for managing data securely and correctly.

Interview Answer

"Data governance ensures compliance, security, ownership, and consistency."

Example

Role-based access for sensitive healthcare data.

Impress Interviewer

"Governance becomes critical in regulated industries like finance and healthcare."

22. Data Quality

What it means

Ensuring data is accurate, complete, and reliable.

Interview Answer

"Poor data quality leads to unreliable analytics and bad business decisions."

Example

Removing duplicate customer records.

Impress Interviewer

"Data observability tools proactively detect anomalies before they impact reporting."

23. Data Lineage

What it means

Tracking data movement from source to destination.

Interview Answer

"Data lineage helps teams understand where data originated and how it changed."

Example

Tracing KPI calculations back to source systems.

Impress Interviewer

"Lineage is essential for debugging and regulatory audits."

24. Data Catalog

What it means

A searchable inventory of organizational data assets.

Interview Answer

"A data catalog improves data discovery and collaboration."

Example

Alation or Collibra cataloging datasets.

Impress Interviewer

"Catalogs improve self-service analytics by making trusted datasets discoverable."

25. Caching

What it means

Storing frequently accessed data temporarily for faster retrieval.

Interview Answer

"Caching reduces latency and database load."

Example

Redis storing frequently used product details.

Impress Interviewer

"Effective caching significantly improves scalability under heavy traffic."

26. Orchestration

What it means

Managing and coordinating workflow execution.

Interview Answer

"Orchestration ensures tasks execute in the correct sequence with dependency handling."

Example

Apache Airflow managing ETL workflows.

Impress Interviewer

"Modern orchestration platforms provide retry logic, alerting, and SLA tracking."

27. Workflow Scheduler

What it means

Triggers jobs based on time or events.

Interview Answer

"A scheduler automates recurring pipeline execution."

Example

Running a daily sales aggregation job at midnight.

Impress Interviewer

"Event-driven scheduling is increasingly replacing purely cron-based scheduling."

28. Fault Tolerance

What it means

Ability of systems to continue working even after failures.

Interview Answer

"Fault-tolerant systems minimize downtime and data loss."

Example

Kafka replicating messages across brokers.

Impress Interviewer

"Replication and checkpointing are common fault-tolerance strategies."

29. Elasticity

What it means

Automatically scaling resources based on demand.

Interview Answer

"Elastic systems scale up during peak load and scale down during low usage."

Example

AWS auto-scaling increasing compute during Black Friday traffic.

Impress Interviewer

"Cloud-native systems separate compute and storage to enable elasticity."

30. Scalability

What it means

Ability to handle increasing workloads.

Interview Answer

"Scalability ensures systems can support business growth efficiently."

Example

Adding nodes to a Spark cluster.

Impress Interviewer

"There's vertical scaling and horizontal scaling, but distributed systems usually prefer horizontal scaling."

GOLDEN INTERVIEW TIPS

When interviewer asks "Difference between ETL and ELT?"

Say:

- ETL transforms before loading.
 - ELT loads raw data first then transforms.
 - ELT is more common in cloud-native analytics.
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Difference between OLTP and OLAP

OLTP	OLAP
Transaction-focused	Analytics-focused
Fast inserts/updates	Complex aggregations
Normalized	Denormalized
Real-time operations	Historical analysis

Difference between Data Lake and Warehouse

Data Lake	Data Warehouse
Raw data	Structured data
Cheap storage	Optimized analytics
Schema-on-read	Schema-on-write
ML & big data	BI reporting

FINAL INTERVIEW ADVICE

Best Strategy

When answering: 1. Define the concept simply. 2. Give a real-world example. 3. Mention one tradeoff. 4. Mention one modern tool.

This makes you sound practical instead of theoretical.

Example Perfect Answer Style

"Kafka is a distributed event-streaming platform used for handling high-throughput real-time data streams. For example, companies use Kafka to process clickstream events from millions of users. One key advantage is scalability and decoupling between producers and consumers."

MOST COMMON TOOLS TO MENTION IN INTERVIEWS

Category	Popular Tools
ETL/ELT	Airflow, dbt, Informatica
Streaming	Kafka, Flink, Spark Streaming
Warehouses	Snowflake, Redshift, BigQuery
Data Lakes	S3, ADLS, GCS
Orchestration	Airflow, Prefect
Catalog/Governance	Collibra, Alation
Caching	Redis
Distributed Compute	Spark, Hadoop

QUICK CLOSING LINE FOR INTERVIEWS

"I focus not only on moving data, but also on making pipelines scalable, observable, and reliable for business decision-making."